

Spectral truncations of the Dirac triple on S^3 : a convergence theorem at explicit rate, a degeneracy obstruction for the obvious Lorentzian extension, and three questions

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This is a three-page summary of two results on Connes–van Suijlekom spectral truncations [1], written to ask three concrete questions of specialists. Verification status is flagged sentence by sentence; proofs are self-contained in the full write-ups, and anything verified only numerically is flagged as such.

1 Setting

Let $\mathcal{T}_{S^3} = (C^\infty(S^3), L^2(S^3, \Sigma), D)$ be the spectral triple of the Dirac operator on the round unit $S^3 = \mathrm{SU}(2)$, with spectrum $\pm(n + \frac{3}{2})$, $n \geq 0$, and multiplicity $(n+1)(n+2)$ per sign. For a cutoff Λ , let P_Λ be the spectral projection and $\mathcal{O}_\Lambda = P_\Lambda C^\infty(S^3) P_\Lambda$ the truncated operator system [1]. Van Suijlekom [2] proved Gromov–Hausdorff convergence of the truncated state spaces $S(\mathcal{O}_\Lambda) \rightarrow S(C(S^1))$ for the circle, and Gaudillot–Estrada–van Suijlekom [3] extended state-space GH convergence to scalar truncations of all compact metric groups — including $\mathrm{SU}(2)$. What the present work adds on the Riemannian side is therefore *not* existence of convergence but: the spinor-bundle (Dirac) substrate, an explicit closed-form rate moment, and the asymptotic rate constant. On the Lorentzian side it adds a negative structural theorem.

2 Construction and rate data

On $\mathrm{SU}(2)$, define the central spectral Fejér kernel

$$K_n(g) = \frac{1}{Z_n} \left| \sum_{j \leq (n-1)/2} \sqrt{2j+1} \chi_j(g) \right|^2, \quad Z_n = \frac{n(n+1)}{2}, \quad (1)$$

a positive, central, unit-mass kernel ($\chi_j =$ trace characters). Its smoothing map $f \mapsto K_n * f$ is unital completely positive (cb-norm 1); its multiplier symbol is given by a Clebsch–Gordan sum

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with $\sigma(0) = 1$, $0 \leq \sigma \leq 1$, support $j' \leq n - 1$. Its mass-concentration moment

$$\gamma_n = \int_{\text{SU}(2)} K_n(g) d_{\text{round}}(e, g) dg = \pi - \frac{4T_n}{\pi Z_n}, \quad T_n = \sum_{\substack{1 \leq k_1, k_2 \leq n \\ k_1 + k_2 \text{ odd}}} \sqrt{k_1 k_2} \left[\frac{1}{(k_1 - k_2)^2} - \frac{1}{(k_1 + k_2)^2} \right], \quad (2)$$

satisfies $\gamma_n \sim \frac{4}{\pi} \frac{\log n}{n}$ — the same constant $4/\pi$ as the classical Fejér–Stein–Weiss circle estimate, here arising as $\text{Vol}(S^2)/\pi^2$ on the Hopf quotient $\text{SU}(2)/U(1)$. A Lipschitz comparison $\| [D, M_f] \| \leq C_3 \|\nabla f\|_\infty$ holds with $C_3 = 1$ (per-harmonic constant $(N - 1)/\sqrt{N^2 - 1}$), and the convolution-form Berezin map $B_n(f) = P_n M_{K_n * f} P_n$ is UCP, an approximate identity at rate γ_n , and Lipschitz-compatible.

3 Result A: unconditional convergence

Theorem 1. *The truncations of the chirality-doubled Dirac triple above converge to $\mathcal{T}_{S^3}$ in van Suijlekom’s state-space Gromov–Hausdorff distance, with the truncated state spaces metrized by the left-translation Lipschitz seminorm $L_n(T) = \sup_{g \neq e} \|U_g T U_g^* - T\|/d(e, g)$ (equal to the metric Lipschitz constant on the continuum): $d_{\text{GH}} \leq \gamma_n = \frac{4 \log n}{\pi n} + O(1/n)$.*

Proof shape (self-contained in the full write-up). The kernel condition $L_n(T) = 0 \Leftrightarrow T \in \mathbb{C}1$ holds on the truthful substrate: left translation preserves harmonic bands, band-limited compression is injective band-by-band (Schur multiplicity-one; the nonvanishing matrix element is the stretched $3\text{-}Y$ coefficient), and transitivity does the rest. The dual map evaluates conjugated operators against the Fejér vector state $\xi = (h \otimes \chi)/\sqrt{Z}$, which lies *exactly* in the Dirac window: under $\text{SU}(2)_L \times \text{SU}(2)_R$ the spinor space is $\bigoplus_j V_j \otimes (V_{j+1/2} \oplus V_{j-1/2})$ and these summands are precisely the Dirac eigenspaces. Both almost-inverse defects collapse to Fejér smoothing — one on functions, one as the operator conjugation-average — at the common moment γ_n , so no partial-inverse or transference estimate appears anywhere. The truthful Dirac-*commutator* seminorm, by contrast, degenerates at finite cutoff (kernel $10/14$ at $n = 2$, $26/55$ at $n = 3$, bit-exact) — it is the wrong finite-cutoff metrization, and this degeneracy is quantified in the write-up. The scalar $C(G)$ pattern is [3]; the contribution here is the spinor-window transport, the per-band injectivity, the closed-form rate moment, and the $4/\pi$ constant.

4 Result B: a degeneracy theorem for the obvious Lorentzian extension

Take the compact-temporal product $S^3 \times S_T^1$ with Krein space $\mathcal{K} = \mathcal{H} \otimes \mathbb{C}^{N_t}$, fundamental symmetry $J = J_s \otimes I$, Lorentzian Dirac $D_L = i(\gamma^0 \otimes \partial_t + D \otimes I)$ (Krein-self-adjoint, van den Dungen’s convention [5]), and the natural operator system of spatial multipliers tensored with momentum-diagonal temporal multipliers. The obvious route to a “Lorentzian quantum metric” is to compress to the Krein-positive eigenspace $\mathcal{K}^+$ of J (where the Krein product is a Hilbert inner product) and run Riemannian machinery on $(P_+ \mathcal{O} P_+, \mathcal{K}^+, P_+ D_L P_+)$.

Theorem 2 (unconditional). *This route is structurally closed. Krein-self-adjointness of the anti-Hermitian spatial block forces $\{J, D \otimes I\} = 0$, hence $P_+(D \otimes I)P_+ = 0$ exactly: the compression annihilates the spatial Dirac. The surviving temporal block is momentum-diagonal and commutes with every momentum-diagonal multiplier. Consequently the compressed-Dirac Lipschitz*

seminorm vanishes identically on the operator system, and no quantum Gromov–Hausdorff-type distance is defined.

Proof sketch. $(iD_s)^\dagger = -iD_s$ with $D_s := D \otimes I$, so $JD_L^\dagger J = D_L$ forces $JD_s J = -D_s$; then $P_+ D_s P_+ = \frac{1}{4}(D_s + D_s J + J D_s + J D_s J) = \frac{1}{4}(D_s + 0 - D_s) = 0$. The temporal commutators vanish since both factors are Fourier-diagonal. Verified bit-exactly at two panel cells (seminorm = 0.0 in `float64` on 42/42 and 275/275 basis multipliers; a frozen regression test guards the statement). $\square$

The failure analysis sharpens the open problem: any viable Lorentzian/Krein convergence construction needs (i) a temporal multiplier algebra the Dirac can see — Toeplitz compressions of e^{iqt} in the Connes–van Suijlekom circle pattern, not momentum-diagonal functions, whose commutators with ∂_t vanish identically; (ii) a Krein-compatible Lipschitz seminorm that does not compress D_L (candidates: seminorms built from JD_L or $|D_L|$, or a genuinely indefinite Lipschitz analog); and (iii) a substrate on which $L(a) = 0 \Leftrightarrow a \in \mathbb{C}1$ actually holds.

5 Three questions

Question 1. *Is the degeneracy theorem (Result B) known — in any form — in the Krein-spectral-triple or Lorentzian NCG literature? It is elementary once stated, which makes prior art likely; we could not locate it.*

Question 2. *For Result A: is the finite-cutoff degeneracy of the Dirac-commutator seminorm — and its repair by the translation seminorm — known or recorded? And is the exact-fit spinor-window embedding of the Fejér vector state (via $V_j \otimes V_{j\pm 1/2}$) the construction you would have used, or is there a slicker route to the dual map on bundle-valued truncations?*

Question 3. *Is the Lorentzian convergence problem, as specified by (i)–(iii) above, of interest to the quantum-metric community — and is anyone already working on a temporal-Toeplitz version?*

References

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